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EE25BTECH11033 - Kavin

Question:
Which one of the following statements is NOT CORRECT?

1) A well-conditioned matrix has a condition number close to 1
2) An ill-conditioned matrix has a large condition number
3) The inverse of a well-conditioned matrix can be computed accurately
4) A non-invertible matrix has a condition number close to 1

Solution:
The **condition number** of an invertible matrix $\mathbf{A}$, denoted as $\kappa(\mathbf{A})$, is defined as:

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \left\|\mathbf{A}^{-1}\right\| \tag{1}$$

where,

$$\|\mathbf{A}\| = \text{norm of matrix } \mathbf{A}$$

It measures how much the output of a linear system $\mathbf{A}\mathbf{x} = b$ can change for a small change in the input data.

In linear algebra, a **matrix norm** is a function that assigns a strictly positive length or size to every non-zero matrix. Conceptually, it is an extension of the idea of a vector norm (which measures the length of a vector) to matrices.

Formal Definition

A function $\|\mathbf{A}\|$ is defined as a matrix norm if it satisfies the following properties for any matrices $\mathbf{A}$ and $\mathbf{B}$ and any scalar c:

1) **Non-negativity:** $\|\mathbf{A}\| \geq 0$, and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A}$ is the zero matrix.
2) **Absolute Homogeneity:** $\|c\mathbf{A}\| = |c| \cdot \|\mathbf{A}\|$.
3) **Triangle Inequality:** $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$.
4) **Submultiplicativity:** $\|\mathbf{A}\mathbf{B}\| \leq \|\mathbf{A}\| \cdot \|\mathbf{B}\|$. (This property is specific to matrix norms).

Common Types of Matrix Norms

There are several different ways to define a matrix norm, with the most common types falling into two categories: induced norms and entry-wise norms.

1. Induced Norms (or Operator Norms)

These norms are defined in terms of how a matrix acts on vectors. An induced norm measures the maximum "stretching factor" that the matrix applies to any vector. It is defined as:

$$||\mathbf{A}||_p = \sup_{\mathbf{x}\neq 0} \frac{||\mathbf{Ax}||_p}{||\mathbf{x}||_p} \tag{2}$$

The most common induced norms are:

- **L1-Norm (Maximum Absolute Column Sum):**

$$||\mathbf{A}||_1 = \max_j \sum_{i=1}^{m} |a_{ij}| \tag{3}$$

- **L-infinity Norm (Maximum Absolute Row Sum):**

$$||\mathbf{A}||_\infty = \max_i \sum_{j=1}^{n} |a_{ij}| \tag{4}$$

- **L2-Norm (Spectral Norm):**

$$||\mathbf{A}||_2 = \sqrt{\lambda_{\max}(\mathbf{A}^*\mathbf{A})} = \sigma_{\max}(\mathbf{A}) \tag{5}$$

  where $\sigma_{\max}$ is the largest singular value of $\mathbf{A}$ and $\mathbf{A}^*$ is the conjugate transpose of $\mathbf{A}$.

2. Entry-wise Norms

These norms treat the matrix as a long vector of its $m \times n$ elements.

- **Frobenius Norm:**

$$||\mathbf{A}||_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2} \tag{6}$$

  This is analogous to the standard Euclidean norm for vectors.

1) **A well-conditioned matrix has a condition number close to 1:** This is **CORRECT**. The smallest possible condition number is 1 (for example, the identity matrix). A value close to 1 indicates that the matrix is numerically stable.

2) **An ill-conditioned matrix has a large condition number:** This is **CORRECT**. This is the definition of an ill-conditioned matrix. A large condition number means that small errors in the input can lead to very large errors in the solution.

3) **The inverse of a well-conditioned matrix can be computed accurately:** This is **CORRECT**. Because a well-conditioned matrix is not sensitive to small perturbations, numerical algorithms used to compute its inverse will produce results with high accuracy.

4) **A non-invertible matrix has a condition number close to 1:** This is **NOT CORRECT**. A non-invertible (or singular) matrix does not have an inverse $\mathbf{A}^{-1}$. For the purposes of conditioning, its inverse is considered to have an infinite norm. Therefore, the condition number of a non-invertible matrix is defined to be **infinity** (∞). An infinite condition number is the largest possible value, representing the worst possible conditioning, not the best (which is close to 1).